

REPORT PhD

EXPERT'S NAME

Prof. Dr. Jianwei Zhang

FILIATION

Faculty of Mathematics, Informatics and Natural Science

Department Informatics, Group TAMS, University of Hamburg

COUNTRY

Germany

EVALUATION

- | | |
|-------------------------------------|---|
| ☒ | PASS |
| ☐ | PASS AFTER REVISION (INCOMPLETE) |
| ☐ | RESUBMIT (CONDITIONAL FAIL) |
| ☐ | DOES NOT PASS |

PhD CANDIDATE'S NAME

Juan Gonzalez-Gomez

TITLE OF PhD THESIS

Modular Robotics and Locomotion: Application to Limbless robots

Place and Date

Signed:

GLOBAL EVALUATION

The last few years have witnessed an increasing interest in modular reconfigurable robotics. The applications include industrial inspection, urban search and rescue, space applications and military reconnaissance. Modular robots are also very interesting for research purposes.

Modular robots are composed of identical or similar units which can interconnect as well as disconnect each other and are capable of changing their configurations.

First, this research on modular robots by the Ph.D. candidate is original and the dissertation is solid. The author presents a completely new, very good classification according to the modular robot topology which is the first of its kind. Then the mathematical, control and kinematical models of controlling modular robots are presented as a background for later discussion. After that, the dissertation concentrates on the locomotion of modular robots in 1D and 2D topologies. The locomotion principles of different configurations are studied thoroughly from the mathematical and kinematics point of view. Different locomotion gaits are investigated originally. More importantly, the dissertation also presents the related on-site experiments to confirm its originality and innovation.

Another very interesting topic in the dissertation is the minimal configurations that modular robotics system requires for locomotion. It is basic and indispensable for understanding the locomotion principle. The author uncovered the number of required for various different motion gaits.

This dissertation answers many unsolved questions of modular robotic research. The reviewer is impressed by the great scientific contribution of this thesis. The topic is approached systemically and the work is well thought-out. First the author investigated the locomotion principle from the mathematical and kinematic point of view. He carried out the necessary simulations to test the new findings. In the end, a cost-efficient modular robot prototype is designed to carry out on-site experiments. The dissertation covers its topic comprehensively and is very well written.

All the achievements of results are definitely important for modular robotic research. A lot of interesting future work is also introduced. The reviewer suggests to the promotion committee that this Ph.D. thesis contributes high quality research and is good enough for a Dr. title.

To evaluate this thesis, please consider the following aspects:

1. - Originality and Innovation.
2. - Planning and Methodology.
3. - Concrete goals and interest in the research area of the thesis.
4. - Interest in the obtained results.
5. - Formal requirements of the thesis and bibliography.